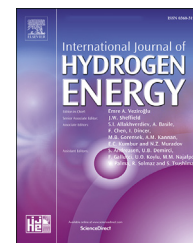


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Measurement of solidification heat from supercooled water freezing during PEFC cold start and visualization of ice distribution

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HIGHLIGHTS

- Two different freezing modes during PEFC cold start are classified.
- Novel and simple method of classifying two modes using sensors was proposed.
- Solidification heat measured by originally developed 6 μm thickness thermocouples.
- Difference of ice distribution inside GDL has been visualized by X-ray CT device.

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ABSTRACT

The water in polymer electrolyte fuel cells (PEFCs) operated below the freezing point interferes with the gas supply, shutting down the operation. Therefore, understanding the freezing mechanism is pivotal to achieving high-performance cold-start. In this paper, a novel method of classifying two different freezing modes using 6 μm thickness thin-film thermocouples (TFTCs) was proposed. We conducted *in situ* measurement of the solidification heat from supercooled water using originally developed TFTCs and further visualized the ice distribution using X-ray CT. In a mode when TFTCs measure rapid temperature rises in the $-10\text{ }^{\circ}\text{C} \sim -5\text{ }^{\circ}\text{C}$, water freezes after diffusing into the GDL, while freezing occurs immediately inside the CL without reaching the GDL in the other mode when TFTCs measure no rapid temperature rise in the $-30\text{ }^{\circ}\text{C} \sim -10\text{ }^{\circ}\text{C}$. These modes should be considered for future researches on components improvement, and optimization of current density sweeping to succeed the cold start.

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Introduction

Polymer electrolyte fuel cells (PEFCs) are used as power generating devices for commercial purposes, such as automobiles or household, owing to their high efficiency and power output [1]. However, PEFCs face various technical challenges, such as cold-start and moisture control [2]. In particular, the voltage drop across the cell due to freezing of water inside the cell is a serious hindrance in realizing high performance in cold environments.

Previous studies have been reported to understand the mechanisms that occur at the cold-start of a cell, achieving a fair degree of success with high-performance cold-starting of PEFCs. Kim et al. [3,4] investigated the effect of a heating and cooling cycle, in the temperature range of -40 – 70 °C, on the components of a PEFC. They reported that the ice formed inside the CL of the PEFC broke the CL structure. Furthermore, the ice formed at the interface of the components created a barrier that separated the components, and consequently increased the proton transport and the ohmic resistance across them. However, these reports are only *ex situ* measurements and do not consider the effects on the actual operation below the freezing point. Guo et al. [5] used cyclic voltammetry to investigate the subfreezing physical properties and performance of a hydrated membrane electrode assembly (MEA) undergoing a heating and cooling cycle. The result revealed that the freezing had caused a decrease in the electrochemical active area, but did not affect the short-term performance of the cell. These investigations further reveal that freezing causes irreversible damage to the cell components.

Generally, water generated by the electrochemical reaction exists as ice below the freezing point. However, at the cold-start of a PEFC, the water exists in the liquid phase, i.e. the supercooled state, for a certain period of time [6]. Therefore, power generation does not terminate at the same time as the startup even at 0 °C, and it can be sustained until the supercooled water has filled the void in the CL. In addition, Tajiri et al. proposed gas purging to remove the water from the CL and gas diffusion layer (GDL) and decide the initial water content of proton exchange membrane (PEM) fuel cell changes the power generation time [7]. The lesser the initial water content, the lesser the open circuit voltage and the longer the power generation time due to the securing of the PEM water storage. Based on these results, Tajiri et al. [8] investigated the process of water evaporation during the purging using an electrochemical measurement method. However, these studies do not specify the local temperature inside the cell.

In other studies, visualization of ice distribution inside the cell has been performed to understand the effect of ice and supercooled water on the performance of the fuel cell. For this visualization, X-ray computed tomography (CT) [9–18], magnetic resonance imaging (MRI) [19–21], and neutron radiography [22–25] under a normal operating temperature range have been used. However, to conduct the visualization at freezing point, a system to cool the cell to subzero temperatures is required. Because introducing cooling systems into these devices is difficult, there have been few reports on the *in situ* visualization at the freezing point. Moreover, in general,

visualizing near the CL with a CT device is difficult because the amount of Pt X-ray absorption is too large. Furthermore, the spatial resolution of neutron radiography is so low that it is difficult to observe the fine ice distribution. Therefore, post-analysis methods, such as CRYO-SEM, to inspect the interiors of the cell after shutdown while maintaining subzero temperatures [26–28] or visualization methods [29–31] using transparent cells are often used. For instance, Tabe et al. visualized ice distribution by observing the cell after shutdown using CRYO-SEM [28]. The result revealed that the freezing mechanism varies with the temperature. It was reported that the product water freezes near the PEM–CL interface at -20 °C and freezes around the GDL at -10 °C [32]. Moreover, Ishikawa et al. [30], using a transparent cell, observed that the generated water deposited on the CL surface had maintained its supercooled state for a certain period of time. This subcooled water freezes when the radius of the water droplets exceeds a specific value, and both the numerical analysis and the experiments demonstrated that this value strongly depended on the degree of subcooling [31]. Furthermore, Oberholzer et al. used neutron imaging to visualize the in-plane directional water distribution and showed that the generated water exists as supercooled water in the range of -20 °C– -10 °C [33]. In addition, they reported the shutdown to have occurred stochastically.

Other researchers investigated the effect of components on the performance of the fuel cell at cold-start. For example, Du et al. studied the effect of the microporous layer (MPL) on the cell performance, and demonstrated the decrease in the humidity dependence of the cell performance and the improvement of cell performance by an anode MPL [34]. Watanabe et al. investigated the influence of GDL pore size on performance, and reported that a GDL with a larger pore size increased the diameter of the ice formation and made it easier for flooding to occur due to the disturbance of the gas by the large ice [35]. Moreover, they presented a new GDL with a three-layer-structure—MPL, carbon paper, and hydrophilic layer. The hydrophilic layer expels the water in the cell, improving the cell performance [36].

Additionally, Ju et al. [37,38] developed a multidimensional transient cold start model to predict cold start behavior and proposed an efficient cold start-up strategy that involves raising the operating current of the PEFC numerically. Furthermore, they showed the effects of CL design parameters on the cold start behavior and provided a novel MPL to enhance the cold start capability of PEFCs using cold start model [39,40].

As mentioned above, research into the cold-start of PEFC has been widely conducted, such as component deterioration analysis, effect of water content of PEM, visualization of ice distribution, component effect determination and cold start model simulation. However, these reports are not sufficient because *in situ* measurements have not been conducted and well-substantiated explanations for the freezing mechanism have not been provided. To the best of our knowledge, no *in situ* temperature measurement method on PEFC cold start exists. Moreover, many experiments have used a cell other than that of the actual devices.

In this study, we succeeded in measuring the local temperature inside the cell during cold start and further

visualized the ice distribution inside the GDL after shutdown occurred. This is achieved by inserting originally developed thin-film thermocouples, which measure the internal temperature during cold startup without affecting the cell performance.

Materials and methods

Cell design and components of polymer electrolyte fuel cell

Fig. 1(a) illustrates the configuration of the small PEFC fuel cell used for this experiment. In addition, as depicted in Fig. 1(b), carbon separators (Special Carbon Co., Ltd.) with five parallel channels were used. The width and depth of all the channels and ribs of the cell were 1 mm, and the reaction area was 1 cm². Additionally, a 30 μ m-thick fluorinated polymer electrolyte membrane was applied on a catalyst-coated membrane (CCM), to ensure water absorption in the proton-exchange membrane (PEM) at cold start. SGL28BC (SIGRACET) was applied on a hydrophobic gas diffusion layer (GDL).

Moreover, cell compression was adjusted by varying the thickness of the black gasket. The optimal gasket thickness was selected by subjecting a GDL to the dry compression test and measuring its thickness after the loading compression using an X-ray computed tomography (CT) device. As a result, a 180 μ m-thick gasket was selected, so that the cell compression would be approximately 1 MPa.

Thin-film thermocouple (TFTC)

In this experiment, a flexible thin-film thermocouple (TFTC) with good mechanical strength was fabricated using micro electromechanical systems (MEMS) technology and inserted into the fuel cell to perform *in situ* measurement of the cell at cold-start. As the thickness of the components of PEFC was approximately 10–100 μ m, wafer-thin thermocouples were used for the direct measurement of the cell without affecting the internal reactions [41]. A schematic of the TFTC is shown

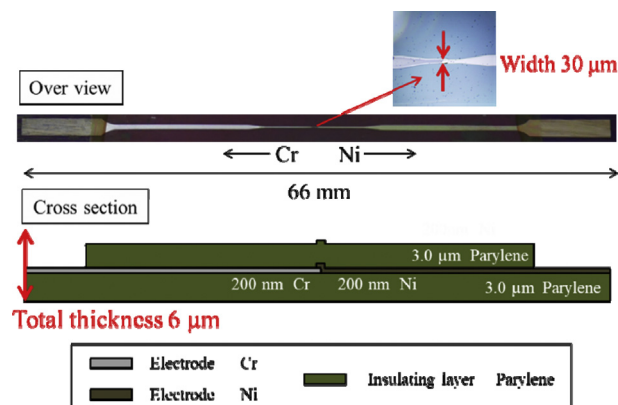


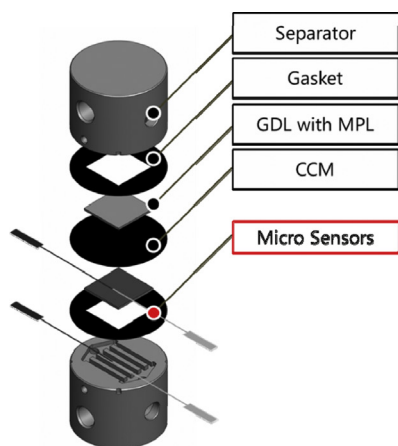
Fig. 2 – Schematic of a thin film thermocouple.

in Fig. 2. The electrodes of the TFTC comprised Ni and Cr. Additionally, a layer of Parylene C, a type of polymer material, was used to insulate the TFTC. The total length, thickness including the insulation layer, and minimum width of the measured part of the sensor were approximately 66 mm, 6 μ m, and 30 μ m, respectively.

Temperature measurement at cold-start

Fig. 3 depicts a schematic of the experimental system used for the *in situ* measurement of solidification heat associated with freezing of the supercooled water. The gas was supplied directly through humidifiers under dry or humid conditions.

The PEFC cold-start experiment was then conducted through the following four steps: a) initial aging, b) purging, c) cooling down, and d) cold start. a) The cell was operated for nearly 1 h. Table 1 lists the operating conditions at each process. b) Because the water freezing in the GDL or CL inhibits the supply of the reaction gas when a PEFC is operated below the freezing point, the residual water inside the cell is “purged.” Furthermore, because the initial moisture condition inside the PEM is also related to its performance at cold-start,



(a) Single cell configuration



(b) Separator

Fig. 1 – Polymer electrolyte fuel cell (PEFC).

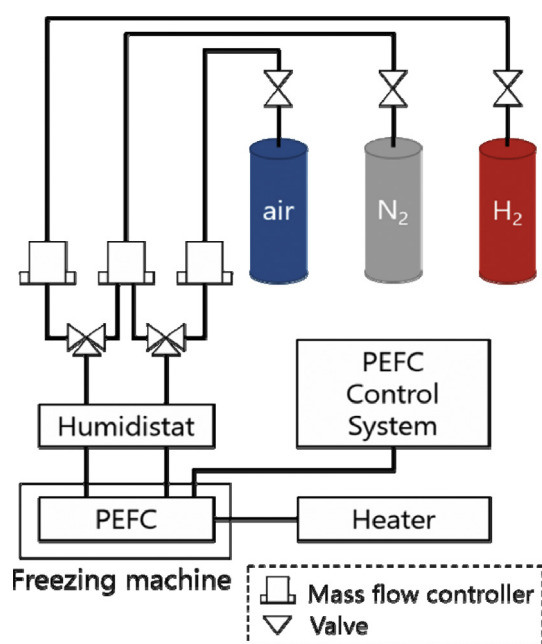


Fig. 3 – Schematic of experimental equipment for measurements.

the moisture conditions in the cell in all the experiments need be unified before operation. Therefore, gas purging was performed after aging.

Gas purging was performed in two stages: 1) dry purging and 2) wet purging. 1) Dry purging with a dry gas (nitrogen, in this case) was performed for 5 min for the purpose of discharging the water generated during the aging operation, and then 2) wet purging with humidified gas at 35% relative humidity (RH) was performed for 90 min to unify the moisture conditions in the cell before the cold-start.

Finally, after this two-stage purging, the cell was placed in a refrigerator (TWINBIRD, SC-DF25), and, after waiting for a steady cell temperature, the cold start operation was performed. The temperature conditions were -5 , -10 , -20 , and -30 °C. The TFTCs were inserted at two positions: cathode CL/MPL interface and GDL/Separator interface. After the shut-down occurred, the cell was moved to the X-ray CT device quickly maintaining subzero temperature and the visualization of ice distribution was conducted.

The cell temperature was measured using a K-type thermocouple (Ninomiya Electric Wire, K-6F) in all the performed experiments. In addition, the current density was constantly controlled at 0.04 A/cm^2 using an electronic load device (KIKUSUI ELECTRONICS, PLZ164W). The cell voltage and TFTC temperature were recorded using a data logger (Yokogawa Electric, MX-100).

X-ray CT visualization

X-ray CT device (Yamato, TDM1000H-II) was used to visualize ice distribution in the cell after shutdown occurred. This CT device was also used in our recent research [41–43]. The X-ray tube voltage was set at 30 kV, the tube current was $100 \mu\text{A}$, whole scanning time was 15 min, and a pixel size was $6.4 \mu\text{m}$.

Results and discussion

Cell voltage and temperature transition at subzero temperatures

Observation of rapid temperature rise inside the cell at -5 °C and verification of solidification heat

Fig. 4 illustrates an example of the transition of cell voltage and measured temperature during cold startup at -5 °C. There are four temperature measurement points: anode separator, cathode separator, CL/MPL interface, and GDL–Sep. interface on the cathode side.

From Fig. 4, the instantaneous cell voltage drop was calculated, with the operation period maintained at -5 °C for several thousands of seconds. Furthermore, TFTCs inserted on the cathode side measured the rapid temperature rise inside the cell at the same time the cell voltage had started to drop. This temperature rise can be measured on both the TFTCs inserted into the CL–MPL interface and the GDL–Sep. interface. Fig. 4(b) depicts an enlarged image of Fig. 4(a) around the time at which the temperature rise is observed. These figures indicate that the TFTC inserted at the CL–MPL interface responds first. On the one hand, a sharp temperature rise was observed at this interface at approximately 1829 s, followed by a gradual drop at the maximum temperature. On the other hand, the temperature rise at the GDL–Sep. interface trailed that at the CL/MPL interface by approximately 1 s.

Table 1 – Experimental conditions.

Initial conditioning	Current density [A/cm^2]	0.5	
	Gas flow rate [mL/min]	Anode	1000
Gas purging		Cathode	1000
	Gas relative humidity [%RH]	100	
	Cell temperature [°C]	50	
	Purging type	Dry purge	
	Time [min]	5	90
Cold startup operation	Gas relative humidity [%RH]	0	35
	Gas flow rate [mL/min]	1000	1000
	Current density [A/cm^2]	0.04	
	Gas flow rate [mL/min]	Anode	6
		Cathode	14
	Gas relative humidity [%RH]	0	
	Cell temperature [°C]	-5 , -10 , -20 , -30	

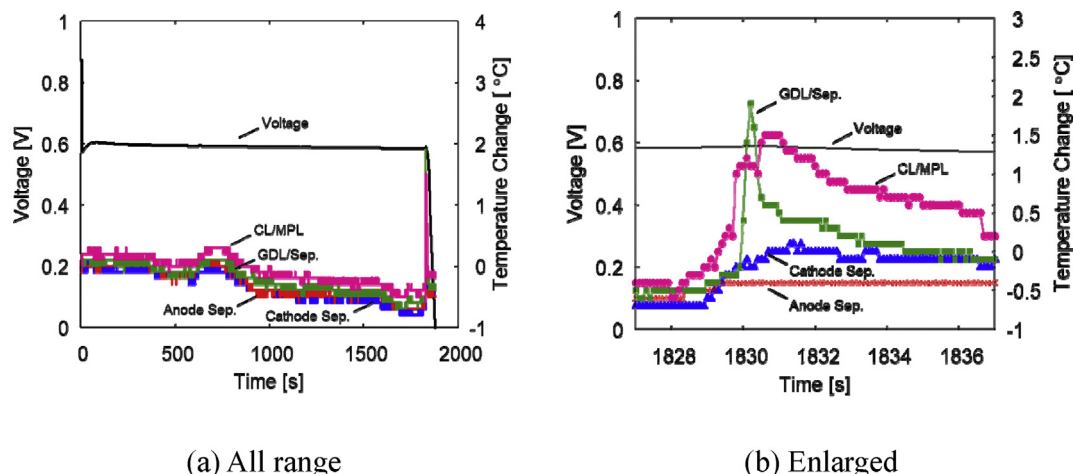


Fig. 4 – Cell voltage and temperature variation during cold-start at -5°C .

Additionally, the temperature increase was approximately 1.9°C at the CL–MPL interface and approximately 2.4°C at the GDL–Sep. interface. Moreover, the temperature of the cathode separator was found to increase by approximately 0.5°C at the same time when the TFTCs gauged the temperature change.

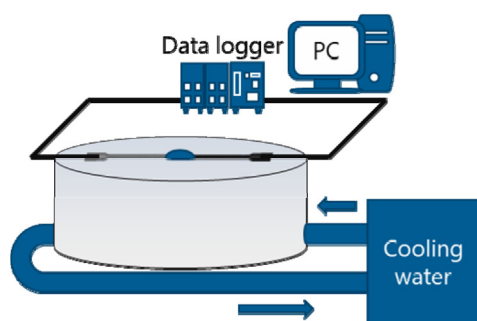
Generally, the degree of subcooling at -5°C is so minute that the water produced inside the cell remains in the supercooled state. In addition, because the exothermic reaction occurs with solidification, the cause of rapid temperature rise at each interface and the cathode separator can be considered as the heat generated when the supercooled water freezes. To confirm this, an experiment for solidification heat measurement was conducted without using PEFC, as depicted in Fig. 5(a). First, a TFTP was placed on the stage and water was poured on its temperature measurement area. Next, the stage was gradually cooled using cooling water. Fig. 5(b) depicts the temperature change of a droplet as indicated by the TFTP at the instant the supercooled droplet freezes. In this measurement, the solidification heat was observed as an instantaneous temperature change in 1 s. Because this behavior is similar to the instantaneous temperature rise

inside the cell during cold-start, the momentary temperature rise of each cathode side interface can be considered as a result of the freezing of water.

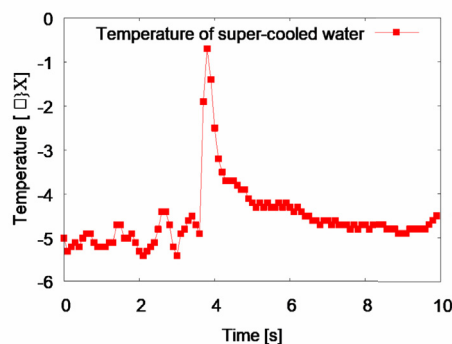
The temperature of each interface and separator increased almost simultaneously with the rapid cell voltage drop. The reason for this voltage drop could be that the ice blocked the supply of reaction gas and caused flooding. Furthermore, because the temperature rise at the CL–MPL interface was the fastest, the origin of the freezing phenomenon was inside the CL or near the TFTP inserted in the CL–MPL interface. Thereafter, the TFTP inserted into the GDL–Sep. interface exhibited a temperature rise. This implies that the TFTP in the GDL–Sep. interface measured the heat transfer from the first solidification or the solidification heat of the supercooled water deposited in the cathode GDL due to heat transfer.

Voltage and temperature variation at -10°C and comparison of temperature change inside the cell at -5°C and -10°C

Fig. 6 illustrates an example of the time transition of cell voltage and measured temperature during the cold startup at -10°C .



(a) Outline



(b) Temperature change of supercooled water freezing

Fig. 5 – Solidification heat measurement without PEFC.

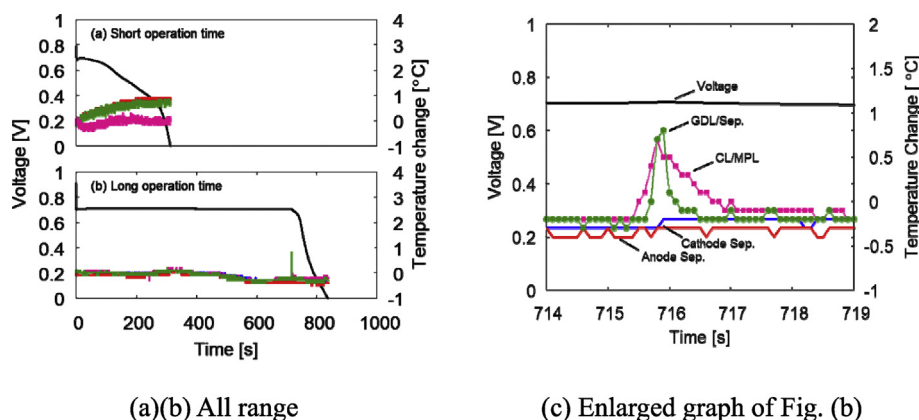


Fig. 6 – Cell voltage and temperature variation during cold-start at -10°C .

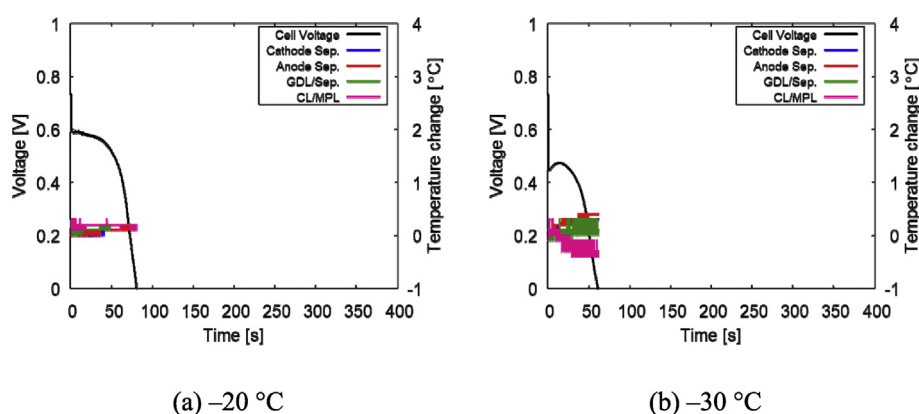


Fig. 7 – Cell voltage and temperature variations during cold-start.

In the -10°C experiments, two trends were observed: a) one with the short operation duration and b) another with the long operation duration. As can be seen from Fig. 6(a), rapid temperature changes inside the cathode components were not observed at the moment when the cell voltage drop occurred (no rapid temperature rise mode). Therefore, this result is different from that of the startup test at -5°C . Thus, it can be concluded that when the long operation at -10°C is impossible, the supercooled water freezes immediately after starting operation inside the CL without reaching the measurement area of the TFTC on the CL/MPL interface. The generated ice inside the CL blocked the gas transportation, and induced shut down. On the other hand, according to Fig. 6(b), a temperature rise was observed as in the case of the fuel cell startup at -5°C (rapid temperature rise mode). Fig. 6(c) depicts the expanded view of Fig. 6(b). This graph shows an instantaneous temperature rise at both the CL–MPL interface and the GDL–Sep. interface. As in the measurement at -5°C , the TFTC inserted in the CL–MPL interface measured the temperature increase earlier, and the TFTC in the GDL–Sep. interface took approximately 0.2 s more to record the temperature increase. However, the absolute temperature increments were approximately 0.9°C at the CL–MPL interface and 1.0°C at the GDL/Sep. interface, respectively, and the temperature increase was smaller than that at -5°C at both the points. This can be attributed to the operating time at

-10°C being much shorter than that at -5°C , and the decrease in the amount of produced water and its calorific value.

Voltage and temperature variation at -20°C and -30°C

No rapid temperature increase was observed at -20°C or -30°C , and the result was similar to the case of the short-duration operation at -10°C . This is attributed to the fact that the water produced during the electrochemical reaction did not diffuse in the temperature measurement area of the TFTCs, and instead instantly became ice under such low temperature conditions. Fig. 7 illustrates an example of the transition of cell voltage and measured temperature at -20°C and -30°C operations, respectively.

Repeatability of cell voltage transition at subzero temperatures

Fig. 8(a) depicts the transition of cell voltage for five iterative measurements conducted at -5°C . The operating durations of all five electric power generations were different. The shortest operating time was approximately 2000 s and the longest time was approximately 12,000 s. The reason that the operating time differs by approximately 10,000 s at the maximum under the same conditions is the probability of freezing of supercooled water. Generally, supercooled water freezes due to the

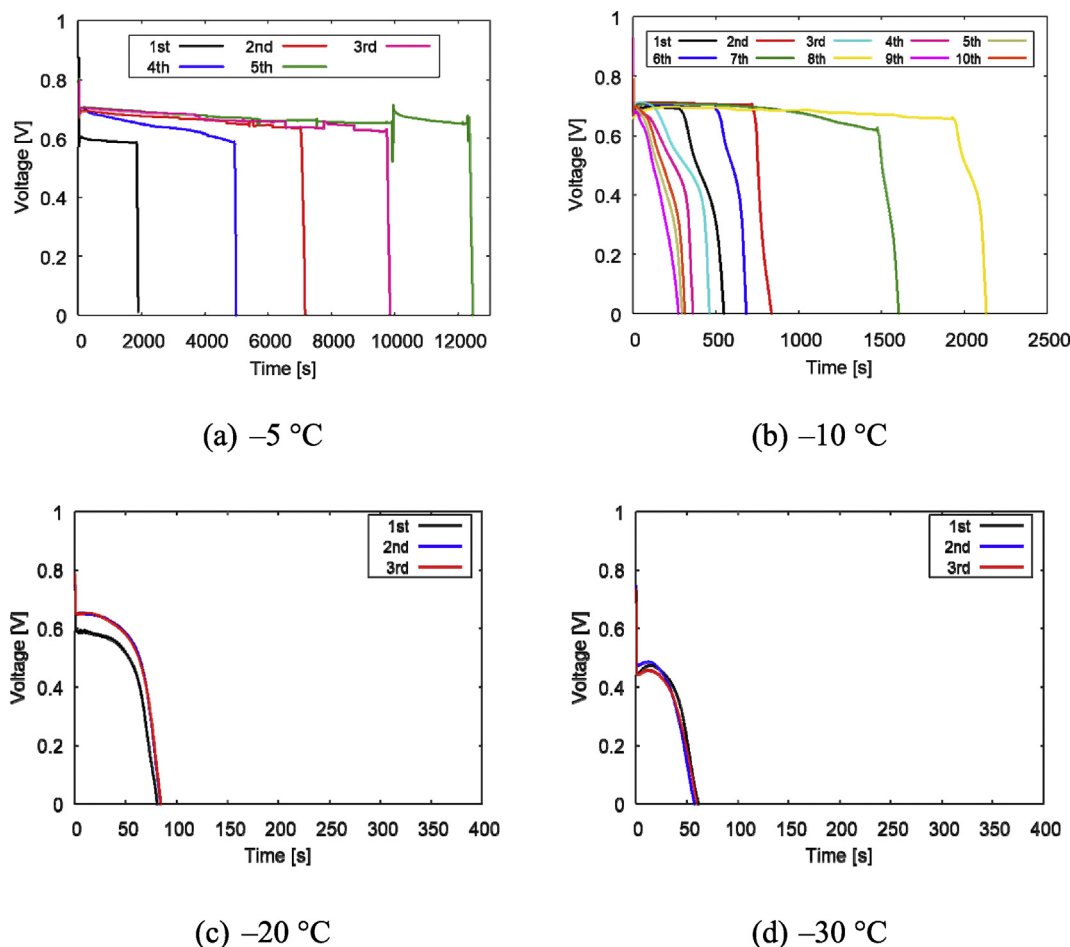


Fig. 8 – Cell voltage transition during cold start at each temperature: (a) $-5\text{ }^{\circ}\text{C}$, (b) $-10\text{ }^{\circ}\text{C}$, (c) $-20\text{ }^{\circ}\text{C}$, (d) $-30\text{ }^{\circ}\text{C}$.

formation of inhomogeneous nucleation, and the occurrence of this nucleation is difficult when the contact angle of the nucleation crystal is large. In this experiment, freezing does not occur easily because of the use of hydrophobic MPL. In addition, because the reaction area of the fuel cell used in this experiment is as small as 1 cm^2 , freezing becomes more uncommon than in a large fuel cell, such as the 25 cm^2 cell used in Ref. [27,44]. This can cause larger variations in the operating time than those in previous studies.

Fig. 8(b) depicts the transition of cell voltage for ten iterations of the experiments during the electric power generation at $-10\text{ }^{\circ}\text{C}$. The TFTCs measured no rapid temperature rise in the four experiments—4th, 5th, 9th, and 10th—that were stopped earlier, while the rapid temperature rise was observed the other six experiments. These results show the existence of two different freezing mode: rapid temperature rise mode and no rapid temperature rise mode during cold start at $-10\text{ }^{\circ}\text{C}$ and this fact is consistent with Schmidt et al. [45]. Although all the operating times varied with the measurement at $-5\text{ }^{\circ}\text{C}$, the difference between the shortest and the longest generation time was approximately 1900 s, which was smaller than that at $-5\text{ }^{\circ}\text{C}$. Furthermore, the operation was mainly terminated in less than 1000 s at $-10\text{ }^{\circ}\text{C}$. Operation was possible only for two iterations—7th and 8th—for more than 1000 s. This result indicates that the operating time of the $-10\text{ }^{\circ}\text{C}$ case is less dependent on the freezing probability of

supercooled water than that of $-5\text{ }^{\circ}\text{C}$. Additionally, the shortest power generation time at $-5\text{ }^{\circ}\text{C}$ and the longest generation time at $-10\text{ }^{\circ}\text{C}$ were approximately the same. From these results, it can be concluded that supercooled water freezes easily at $-10\text{ }^{\circ}\text{C}$ than at $-5\text{ }^{\circ}\text{C}$.

Fig. 8(c) depicts the transition of cell voltage during its operation at $-20\text{ }^{\circ}\text{C}$. The experiments were conducted thrice. The operating duration was approximately 80 s for all measurements, which was shorter than the $-10\text{ }^{\circ}\text{C}$ experiment. This is because freezing could easily occur with the increase in the degree of supercooling. Moreover, the transition of cell voltage was approximately the same in all the experiments undertaken. This result indicates that freezing at $-20\text{ }^{\circ}\text{C}$ has little influence on the freezing probability of supercooled water.

Fig. 8(d) depicts the transition of cell voltage during power generation at $-30\text{ }^{\circ}\text{C}$. Similar to the $-20\text{ }^{\circ}\text{C}$ experiment, the change in cell voltage was approximately the same in all experiments. The operating duration was short at approximately 60 s.

As mentioned, during the shutdown at low temperature conditions, such as $-20\text{ }^{\circ}\text{C}$ and $-30\text{ }^{\circ}\text{C}$, the water produced by the electrochemical reaction is instantly frozen inside the CL and is not transported outside the layer. Therefore, it can be concluded that the freezing probability of supercooled water is not expressed in such low temperature conditions.

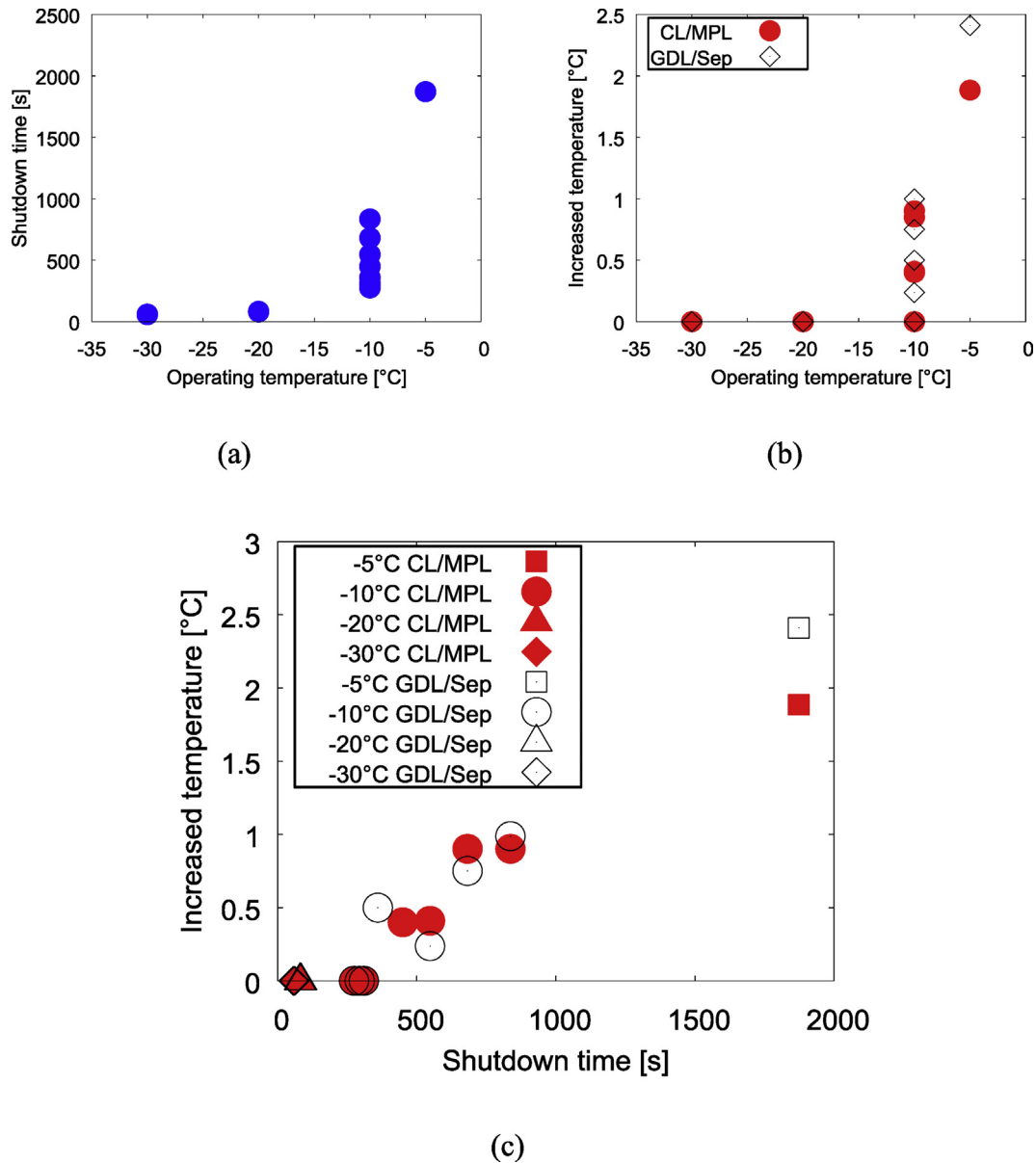


Fig. 9 – Correlation of values shown on each axis: (a) operating temperature and duration, (b) operating and increased temperature, (c) operating duration and increased temperature.

Correlation between operating duration and temperature rise, and calculation of generated water during power generation

As these results indicate, TFTCs can measure the latent heat released during the solidification at shutdown. Fig. 9(a) depicts the correlation between the operating temperature and the generating duration, whereas Fig. 9(b) depicts the relationship between the operating temperature and the temperature increase measured by the TFTCs.

Fig. 9(a) indicates that the water generation duration tends to be shorter due to the lower operating temperature. This is because, as the degree of supercooling increases, the generated water tends to solidify, and the ice causes flooding and forcibly terminates the operation. However, at subzero temperatures

Table 2 – Volume of produced water during rehydration period and after rehydration.

Existence of rapid temperature rise	Op. temp. [°C]	Duration of hydration (upper) and after hydration (lower) [s]	Volume of produced water [mm ³]
Rapid	–5	87 1786	0.32 6.7
Rapid	–10	96 740	0.36 2.8
No	–10	20 294	0.075 1.1
No	–20	10 73	0.037 0.27
No	–30	13 48	0.049 0.18

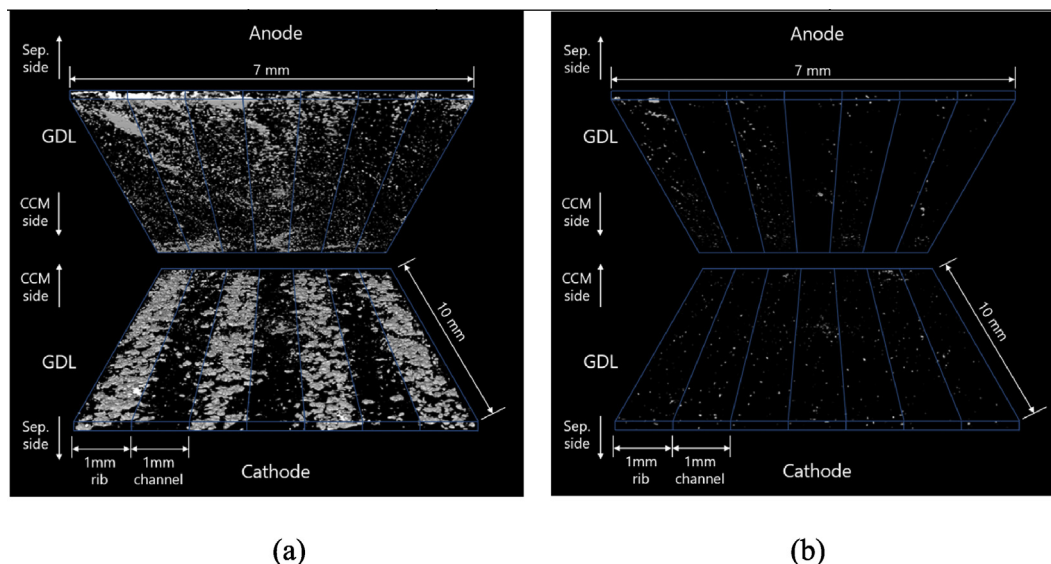


Fig. 10 – 3D images of the ice distribution inside the GDL visualized by X-ray CT: (a) in the rapid temperature rise mode at $-5\text{ }^{\circ}\text{C}$, (b) in the no rapid temperature rise mode at $-20\text{ }^{\circ}\text{C}$.

higher than $-10\text{ }^{\circ}\text{C}$, it is easier for water to exist in a supercooled state, which leads to the higher dependence of the probability of freezing of supercooled water on shutdown. Hence, these facts make the operating duration vary. In fact, Fig. 8(a) and (b) reveal a variation in the generating time at $-5\text{ }^{\circ}\text{C}$ and $-10\text{ }^{\circ}\text{C}$.

Fig. 9(b) depicts the temperature increase at the CL–MPL interface and the GDL–Sep. interface with the operating temperature. This trend may be attributed to the increase in the amount of supercooling water in the cell with the generating duration.

Fig. 9(c) depicts the relationship between the temperature increase and the operation time. The red plots are the values at the CL–MPL interface and the white plots are the values at the GDL–Sep. interface. These figures indicate the increase in the temperature increments with the prolonged operating duration at both interfaces. This is because the total amount of latent heat increased as the amount of frozen water increased.

Cell area occupied by freezing water and ice distribution from calculation of produced water and X-ray CT visualization

The amount of produced water during power generation was calculated by multiplying the rate of water production by the cathode reaction and the operating duration as

$$m_{\text{H}_2\text{O}} = i/2F \times M_{\text{H}_2\text{O}} \times t \quad (1)$$

where i is the current density, F is Faraday's constant, $M_{\text{H}_2\text{O}}$ is the molecular weight of water, and t is the duration. The calculated amount of generated water was divided into two parts: the amount of water during the membrane rehydration and that after rehydration. The amount of water removed by the reactant gases was negligible and hence, was not considered. The period of the membrane rehydration was defined as the period when the cell voltage was increasing [46]. The volume of the produced water presented in Table 2 was obtained by

dividing the mass of the produced water during each period by the density of the water. According to Table 2, the volumes of water during the hydration period at $-5\text{ }^{\circ}\text{C}$ and $-10\text{ }^{\circ}\text{C}$ in the rapid temperature rise mode were 0.32 and 0.36 mm^3 , respectively, which were the amount of water absorbed by the membrane. Additionally, the volumes of water after hydration at $-5\text{ }^{\circ}\text{C}$ and $-10\text{ }^{\circ}\text{C}$ were 6.7 and 2.8 mm^3 , respectively, which were the amount of water diffused into the GDL. However, the volume of hydrated water at $-10\text{ }^{\circ}\text{C}$, $-20\text{ }^{\circ}\text{C}$, and $-30\text{ }^{\circ}\text{C}$ in the no rapid temperature rise mode was much lesser than the amount of water that the membrane can hydrated and shutdown occurred before the membrane completely absorbed it. Moreover, in these conditions, the total amount of produced water was lesser than the volume of the CL (1.5 mm^3). Therefore, it is difficult for the water to reach the GDL, which is considered to be the reason why TFCs could not observe the temperature rise. Furthermore, 3D images of the ice distribution inside the GDL obtained by X-ray CT device are presented in Fig. 10. Fig. 10(a) was taken after power generation in the rapid temperature rise mode at $-5\text{ }^{\circ}\text{C}$ and (b) was in the no rapid temperature rise mode at $-20\text{ }^{\circ}\text{C}$. Inside the cathode GDL, ice was observed in Fig. 10(a), while there was no ice in Fig. 10(b). Additionally, Fig. 10(a) shows that the amount of ice in the cathode GDL under the rib is larger than the amount of ice under the channel. This distribution of ice has the same tendency as the distribution of liquid water in the cathode GDL during PEFC power generation at temperature above $0\text{ }^{\circ}\text{C}$, such as room temperature [47,48]. Therefore, in the rapid temperature rise mode such as freezing mode at $-5\text{ }^{\circ}\text{C}$, the generated water is considered to be transported from the CL to the GDL as supercooled water or condensed in the GDL component and then frozen.

From these results, it can be concluded that the distribution of the supercooled water inside the cell largely depends on the operating temperature and operating duration. If the degree of subcooling is relatively small, such as $-5\text{ }^{\circ}\text{C}$ or

–10 °C, and a freezing mode is rapid temperature rise mode, the supercooled water diffuses in the GDL or cathode separator, and then freezes. However, if the temperature is extremely low, such as –20 °C or –30 °C, and a freezing mode is the no rapid temperature rise mode, the produced water freezes immediately inside the CL.

Conclusions

The PEFC cold start experiments were performed while measuring the temperature of cathode CL/MPL interface and GDL/Separator interface. The operation temperatures were –5, –10, –20, and –30 °C. As a result, the power generating duration tends to be shorter due to the lower operating temperature since the generated water tends to solidify, and the ice causes flooding and forcibly terminates the operation as the degree of supercooling increases. However, at subzero temperatures higher than –10 °C, it is easier for water to exist in a supercooled state, which leads to the higher dependence of the probability of freezing of supercooled water on shutdown. Additionally, the sharp temperature rise at the CL–MPL interface and the GDL–Sep. interface increases with the operating temperature due to the increase in the amount of supercooled water in the cell with the power generating duration.

There are two different freezing modes: the rapid temperature rise mode and the no rapid temperature rise mode during PEFC cold start. In the rapid temperature rise mode in the –10 °C ~ –5 °C, the water generated by the electrochemical reaction remained in the supercooled state, and froze after it reached the GDL or separator. On the other hand, in the no rapid temperature rise mode in the –30 °C ~ –10 °C, water froze immediately after the start of the operation inside the CL without reaching the GDL.

Distinguishing these two freezing modes is significant to optimize the properties of components, such as GDL and MPL, and improve the current density sweep method to succeed the cold start. In this paper, the solidification heat from supercooled water freezing during the PEFC cold-start was measured using originally developed 6 µm thickness thin-film thermocouples (TFTCs) and the visualization of ice distribution using X-ray CT was conducted. Results show that the *in situ* measurement of solidification heat using TFTCs can be a novel and simple method to classify two freezing modes during cold start.

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